

European Option Pricing & Risk Toolkit

Black–Scholes–Merton Analytical Engine

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Parameter	Value
Spot (S)	150.0000
Strike (K)	155.0000
Time to expiry (T)	0.5000 years
Risk-free rate (r)	5.0000%
Volatility (σ)	25.0000%
Dividend yield (q)	0.0000%
Moneyness S/K	0.9677

This report documents closed-form BSM prices, full Greeks, Monte Carlo cross-validation with standard errors, implied-volatility inversion (Brent), and put–call parity diagnostics for a European equity option.

1. Model & Methodology

Under the Black–Scholes–Merton assumptions the underlying follows geometric Brownian motion with constant drift and volatility. In the risk-neutral measure:

$$dS_t = (r - q) S_t dt + \sigma S_t dW_t^Q$$

The unique no-arbitrage price of a European claim is the discounted risk-neutral expectation of its payoff. Closed forms:

$$d1 = [\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T] / (\sigma\sqrt{T}), \quad d2 = d1 - \sigma\sqrt{T}$$

$$\text{Call} = S e^{-qT} N(d1) - K e^{-rT} N(d2)$$

$$\text{Put} = K e^{-rT} N(-d2) - S e^{-qT} N(-d1)$$

Put–call parity: $C - P = S e^{-qT} - K e^{-rT}$. Monte Carlo uses exact GBM terminal sampling with antithetic variates. Implied volatility is solved with Brent’s root finder on the map $\sigma \mapsto \text{BSM}(\sigma) - \text{market price}$.

2. Pricing Results

Metric	Call	Put
BSM analytical price	10.029857	11.202893
Intrinsic value	0.000000	5.000000
Time value	10.029857	6.202893
Monte Carlo price	10.062091	11.217552
MC standard error	0.037824	0.032267
MC 95% CI	[9.9880, 10.1362]	[11.1543, 11.2808]
MC – BSM	0.032234	0.014659
Implied vol (from BSM price)	25.000000%	—

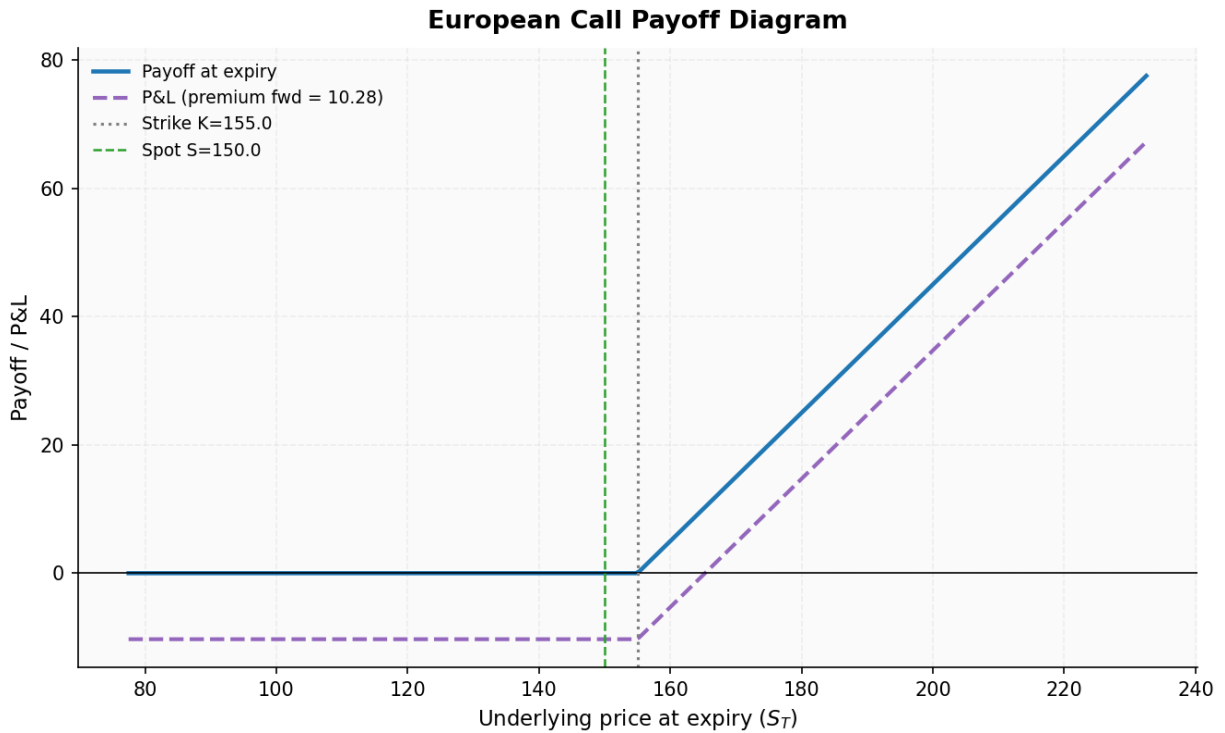
Put–call parity residual = -1.42e-14 (HOLDS at tol=1e-08). $C - P = -1.173036$, $S e^{-qT} - K e^{-rT} = -1.173036$.

3. Risk Sensitivities (Greeks)

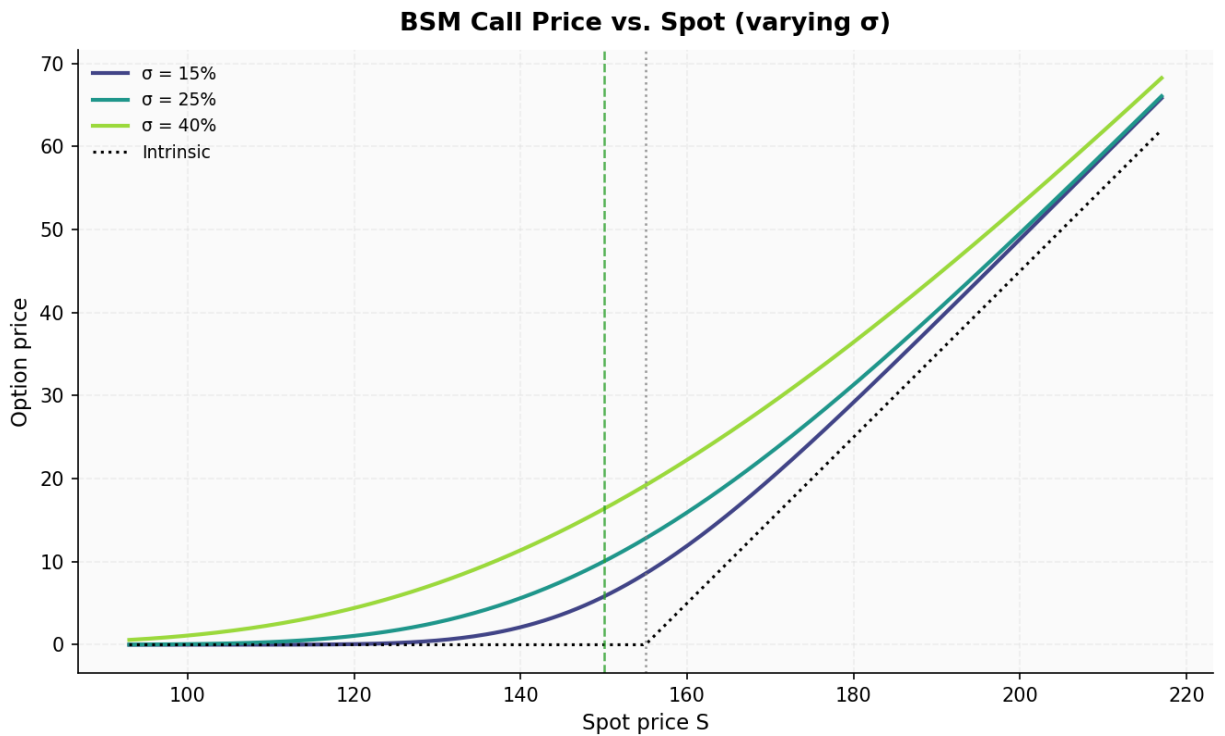
Greek	Call	Put	Interpretation
Delta (Δ)	0.517676	-0.482324	$\partial V / \partial S$ — hedge ratio
Gamma (Γ)	0.015030	0.015030	$\partial^2 V / \partial S^2$ — convexity
Vega (v) / 1%	0.422727	0.422727	$\partial V / \partial \sigma$ per vol point
Theta (Θ) / day	-0.038217	-0.017508	Time decay per day
Rho (ρ) / 1%	0.338108	-0.417757	$\partial V / \partial r$ per rate point
Vega (absolute)	42.272676	42.272676	$\partial V / \partial \sigma$ (σ in decimal)
Theta (per year)	-13.949249	-6.390597	$\partial V / \partial t$ annualized

4. Visual Analytics

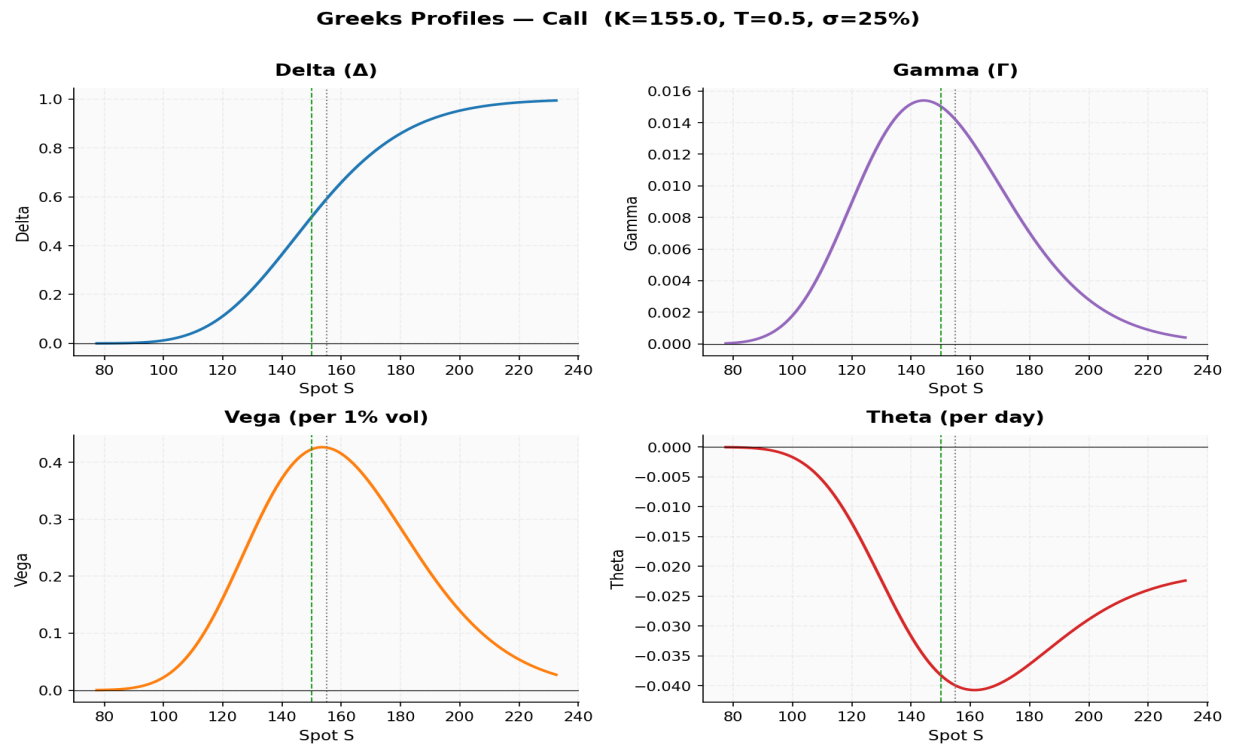
Payoff & P&L at Expiry



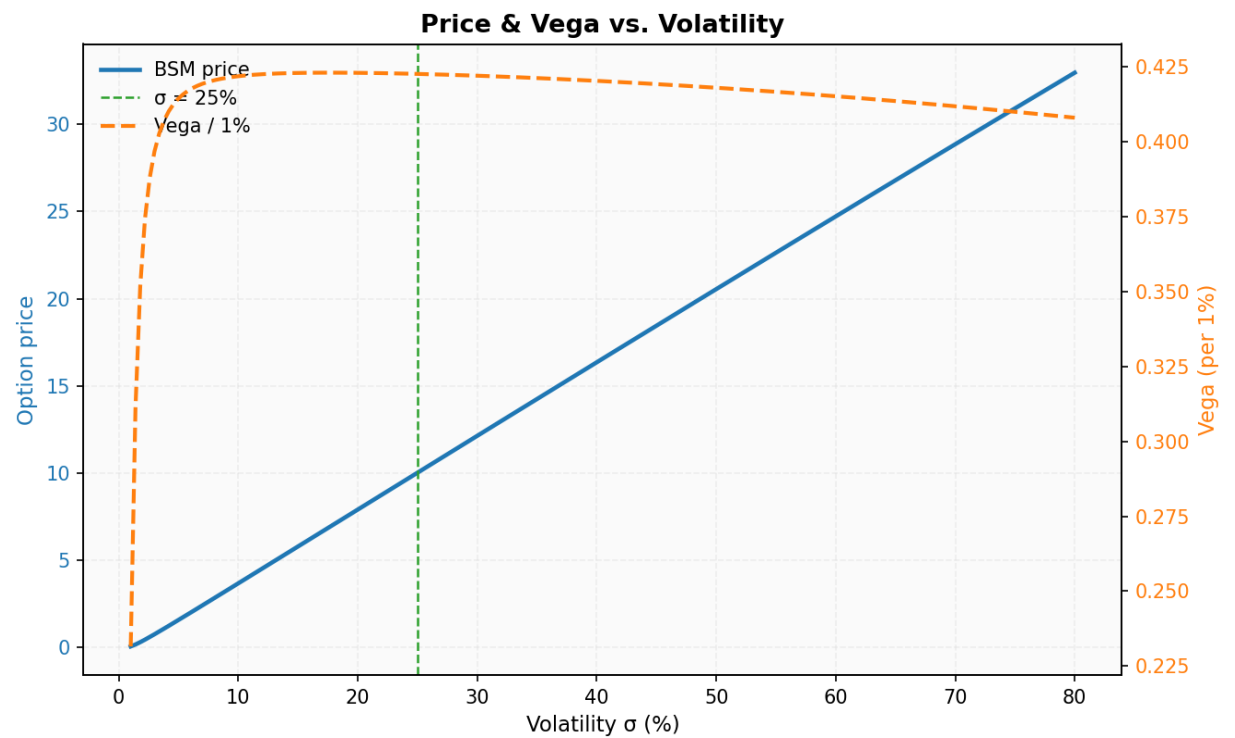
Price vs. Spot for Varying Volatilities



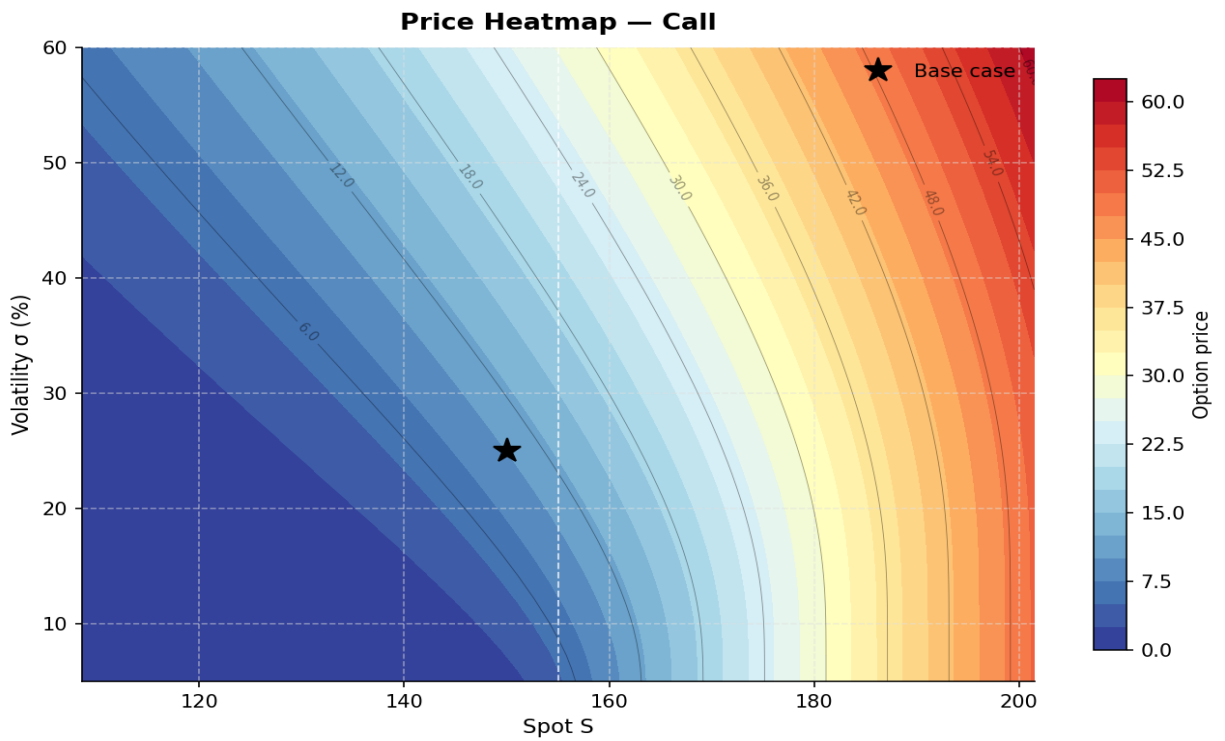
Greeks Profiles vs. Spot



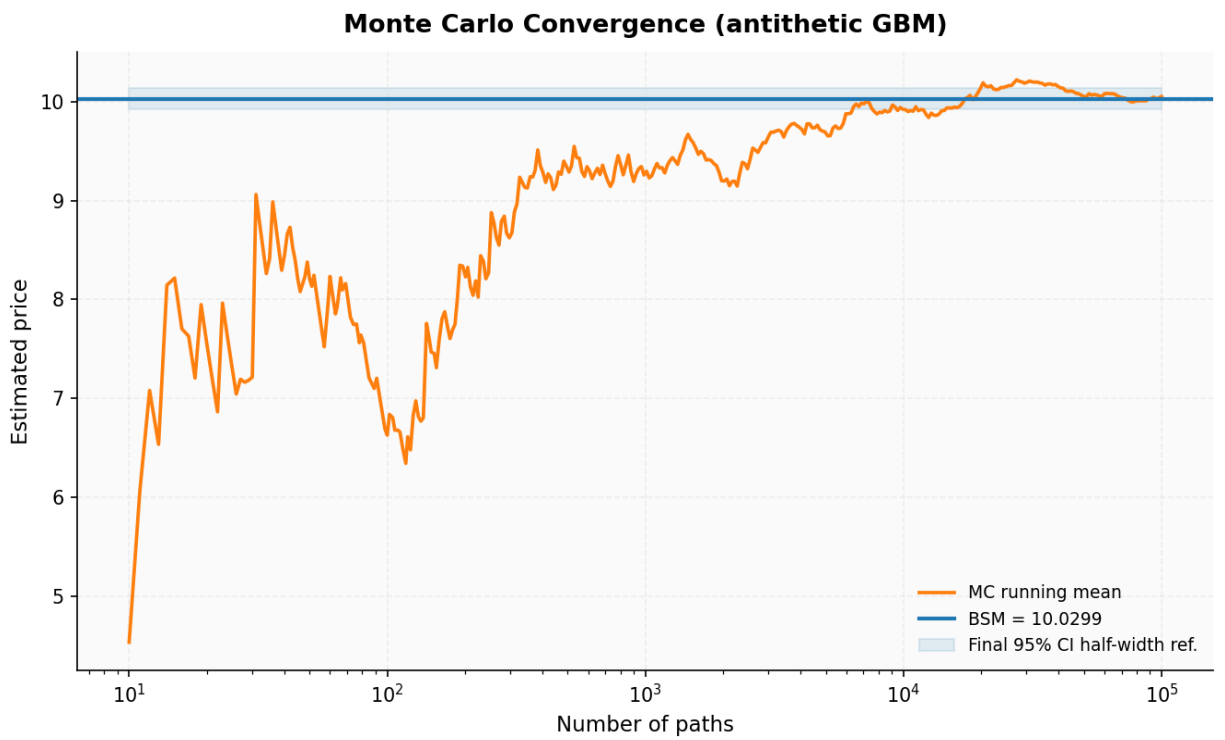
Price & Vega vs. Volatility



Price Heatmap ($S \times \sigma$)



Monte Carlo Convergence



5. Summary & Extensions

For the base case ($S=150.0$, $K=155.0$, $T=0.5$, $r=0.05$, $\sigma=0.25$) the European call is valued at **10.0299** and the put at **11.2029**. Monte Carlo (antithetic, $n \approx 200,000$) recovers both prices within sampling error. Implied volatility from the model price round-trips to the input σ (25.0000%), confirming solver integrity. Parity residual is numerically zero ($-1.42e-14$).

Natural extensions: American options via CRR binomial or finite-difference PDE; local/stochastic volatility; FX Garman–Kohlhagen (set q = foreign rate); barrier / Asian path-dependent MC; SABR smile calibration; portfolio-level Greeks and scenario P&L.